

INSIGHTS FROM THE DATA: Online Advertising Performance Data

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FINLATICS

A solid orange horizontal bar spanning the width of the slide, located at the bottom.

What is the overall trend in user engagement throughout the campaign period?

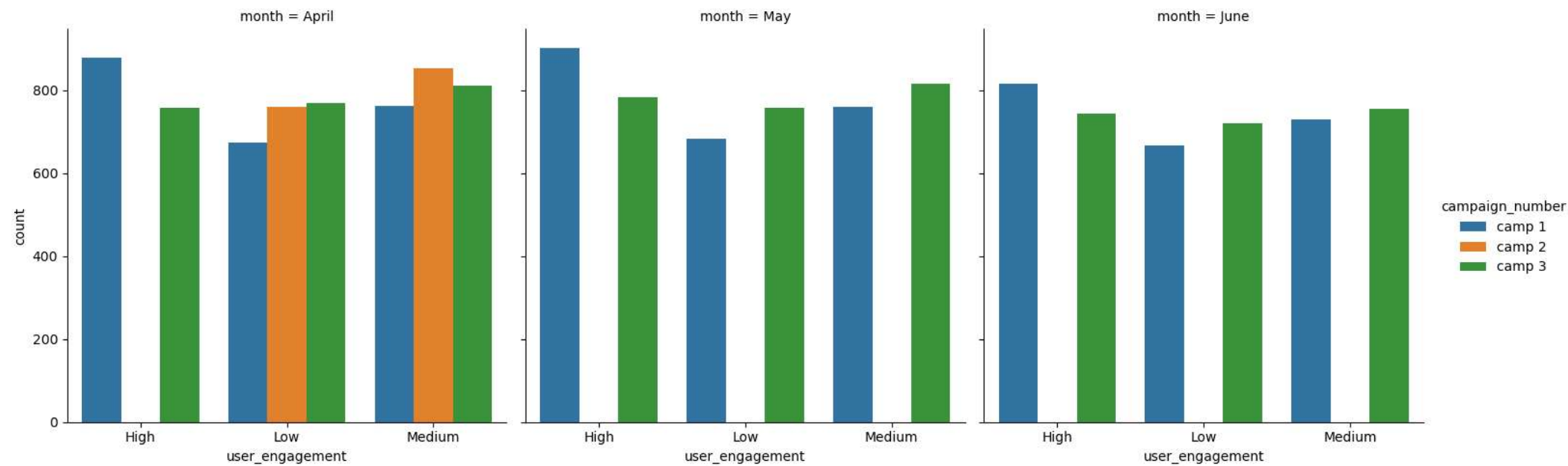
user_engagement

Medium 5489

Low 5035

High 4884

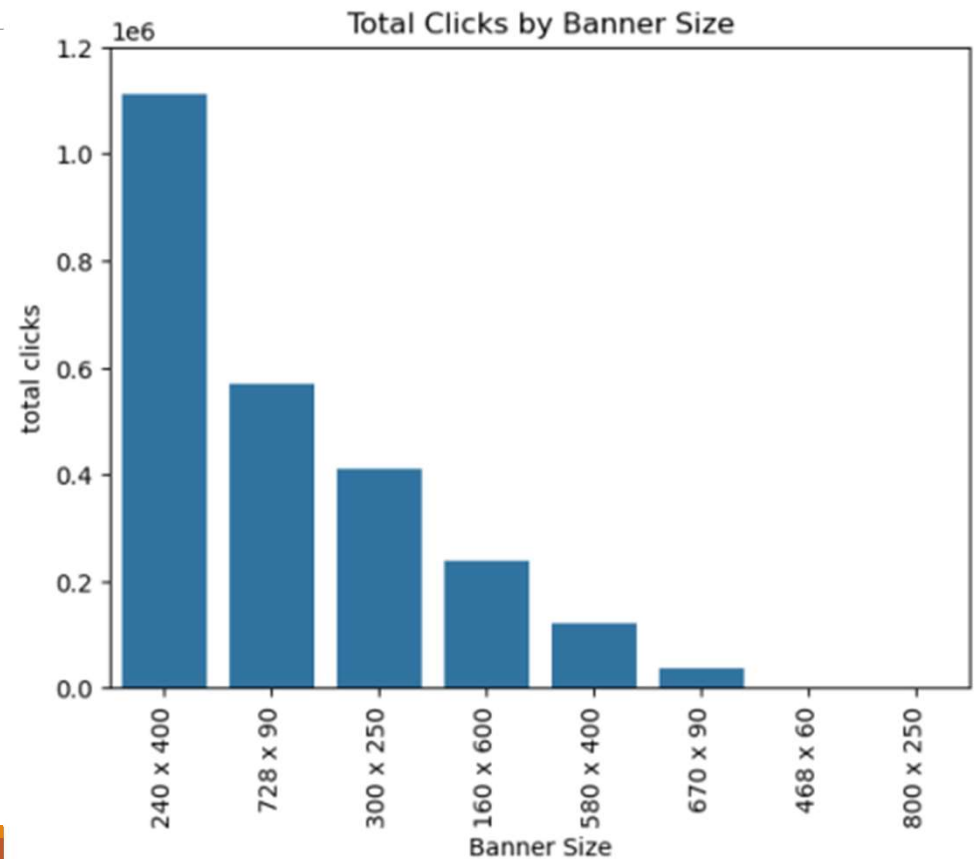
Month	Campaign	High	Medium	Low
April	Camp 1	879	762	674
	Camp 2	–	853	761
	Camp 3	758	812	770
May	Camp 1	903	760	683
	Camp 3	784	816	759
June	Camp 1	817	731	668
	Camp 3	743	755	720



Camp 2 was only there in April. It shows NO high engagement. Camp 1 performs reasonably better throughout the period. Though High engagement is in camp one in every month, in lower and medium engagement camp 3 is more in number

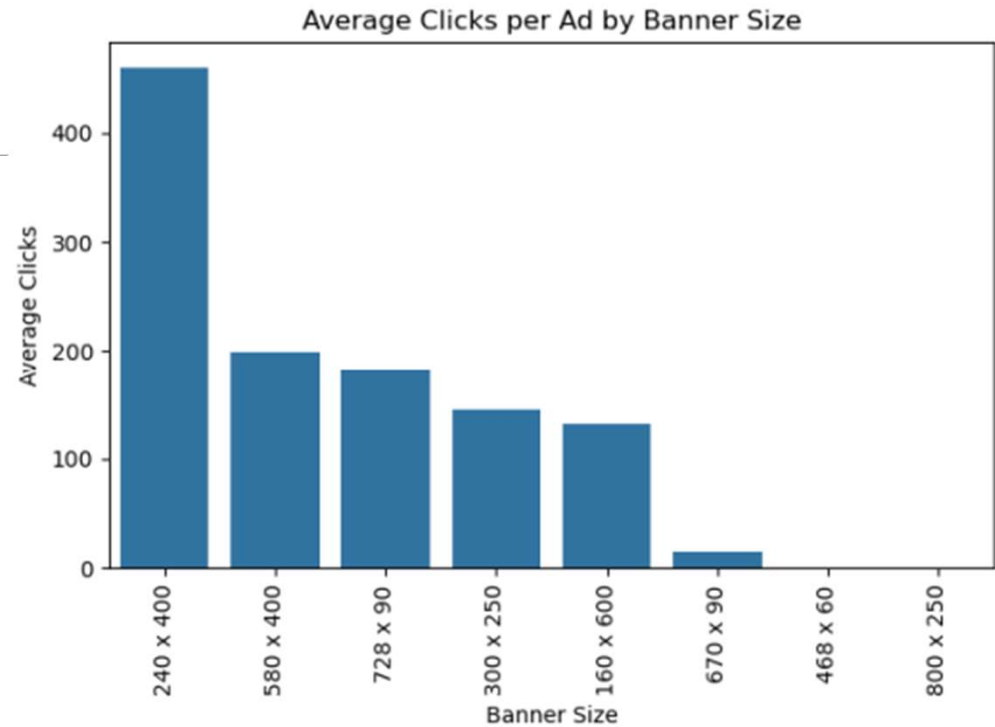
How does the size of the ad (banner) impact the number of clicks generated?

Banner	total clicks
160 x 600	239570
240 x 400	1113256
300 x 250	411214
468 x 60	1295
580 x 400	120681
670 x 90	37203
728 x 90	569606
800 x 250	12



The banner 240X400 has the greatest number of clicks and the greatest number of average clicks.

800x250 is very poorly performing as clearly shown



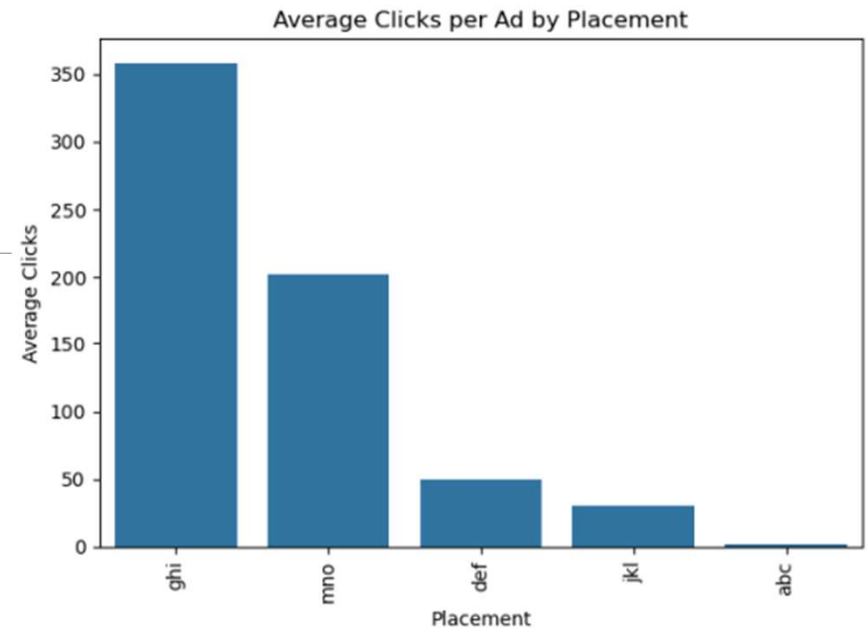
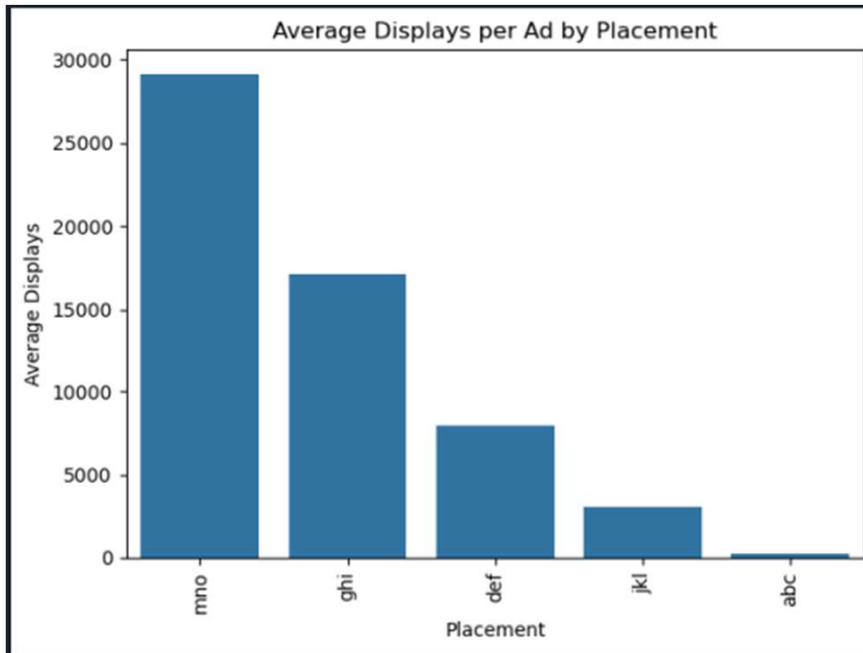
Which publisher spaces (placements) yielded the highest number of displays and clicks?

Placement	Total Click
ghi	1247049
mno	993044
def	176097
jkl	75063
abc	1584

Placement	Total displays
mno	143164944
ghi	59740415
def	28177492
jkl	7692732
abc	242142

Placement	Average Clicks
ghi	357.935993
mno	202.084656
def	49.773036
jkl	29.977236
abc	1.636364

Placement	Average Displays
mno	29134.095238
ghi	17147.076636
def	7964.243075
jkl	3072.177316
abc	250.146694



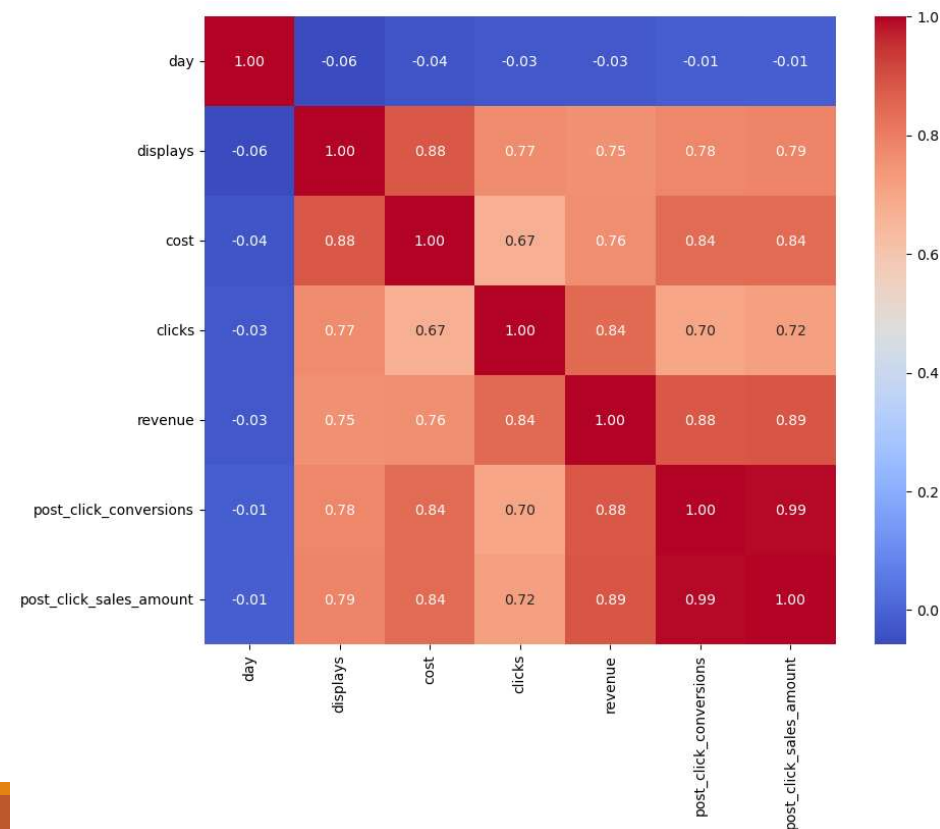
While the Average and total clicks generated by placement GHI is higher, the average and total Displays generated of MNO is higher. ABC placement performs poorly in both. The mode of the placements is Also MNO.

Is there a correlation between the cost of serving ads and the revenue generated from clicks?

CLEARLY SHOWS HIGH CORRELATION.

~0.67

If the cost of serving ad is more,
Then the revenue generated through it
is also higher.



What is the average revenue generated per click for Company X during the campaign period?

Overall Average Revenue per Click:
0.1108

Which campaigns had the highest post-click conversion rates?

campaign_number	Post click conversions
-----------------	------------------------

camp 1	633085
--------	--------

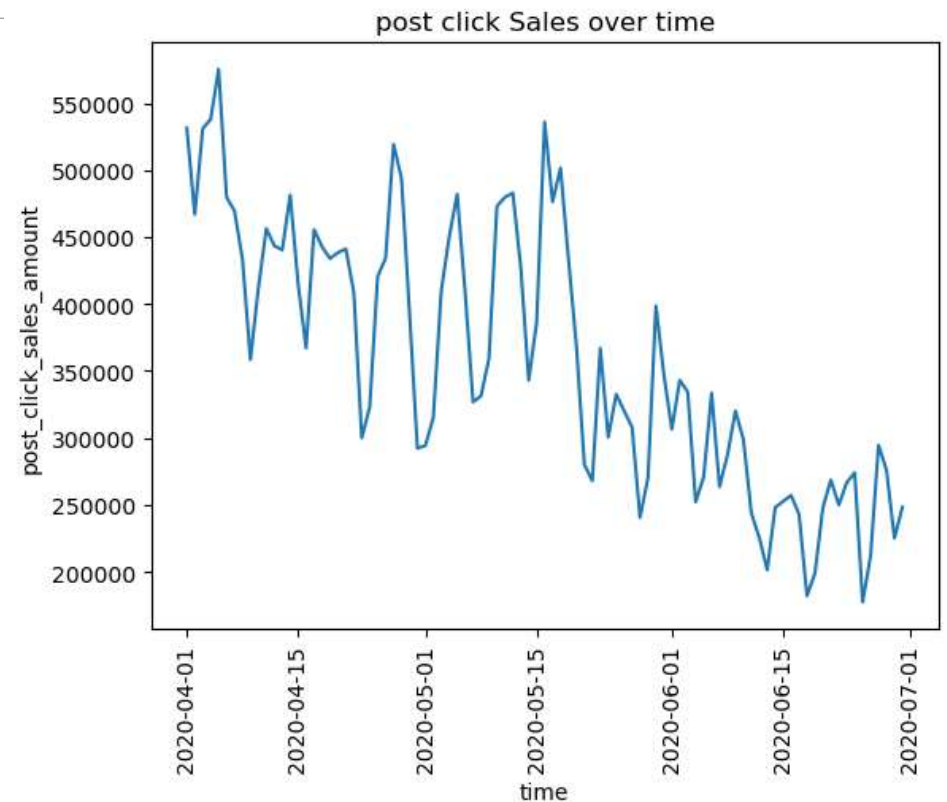
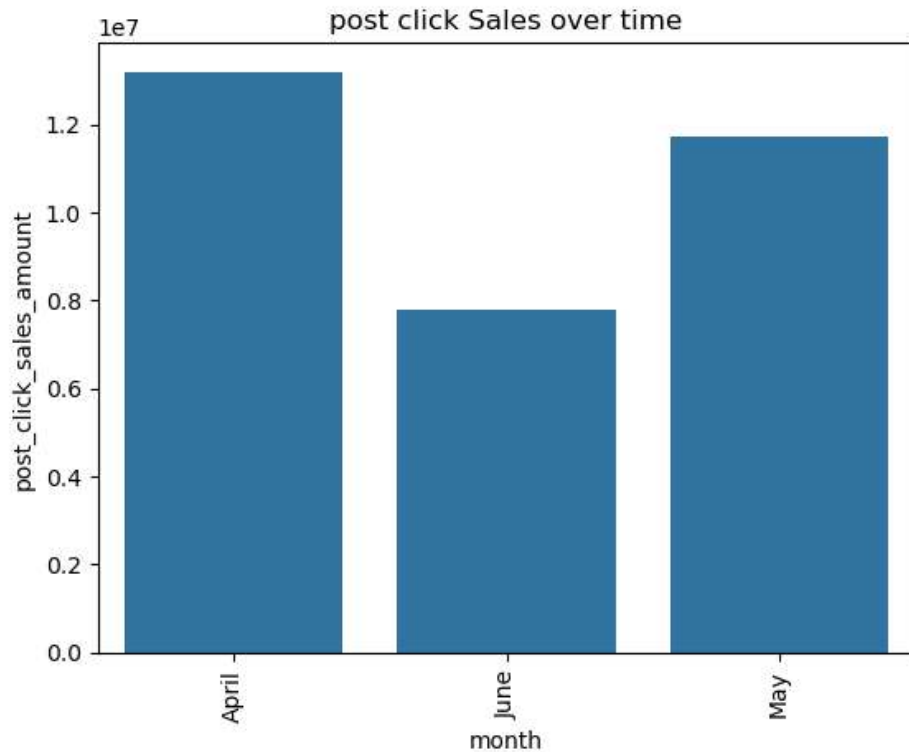
camp 2	13767
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camp 3	4916
--------	------

camp 1 has most post click conversions

CAMP 1 Outperforms all other camps in many aspects as will be seen later

Are there any specific trends or patterns in post-click sales amounts over time?



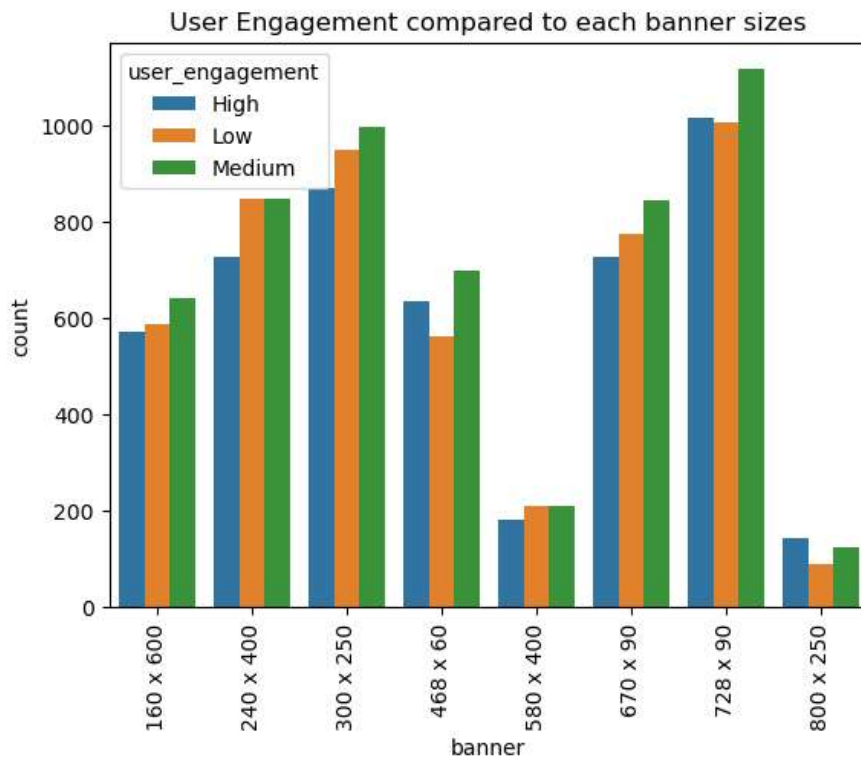
The Sales are higher in the months of April and May and decreases in June.

The Post click Sales decreases over time. We see a regular rise and fall in the sales though-out the time period.

Overall Sale does decrease. Till 15th of may the maxima are quite near but after that they too fall



How does the level of user engagement vary across different banner sizes?



Banner Size	High	Medium	Low
160 × 600	573	642	590
240 × 400	729	848	848
300 × 250	872	999	949
468 × 60	637	699	563
580 × 400	182	212	212
670 × 90	728	847	777
728 × 90	1018	1117	1007
800 × 250	145	125	89

The most engagement is with the 728X90 Banner. Followed by 300x250. Even in number 720X90 is most. The 800x250 Banner is the least engaged among the banners.

Number of entries for each banner

banner	
728 x 90	3142
300 x 250	2820
240 x 400	2425
670 x 90	2352
468 x 60	1899
160 x 600	1805
580 x 400	606
800 x 250	359

Which placement types result in the highest post-click conversion rates?

Placement GHI has the most post-click conversion rates even though MNO is the mode of the data

```
Placements by post click conversion rates
```

```
placement
```

```
ghi      337063
```

```
mno      263172
```

```
def       29856
```

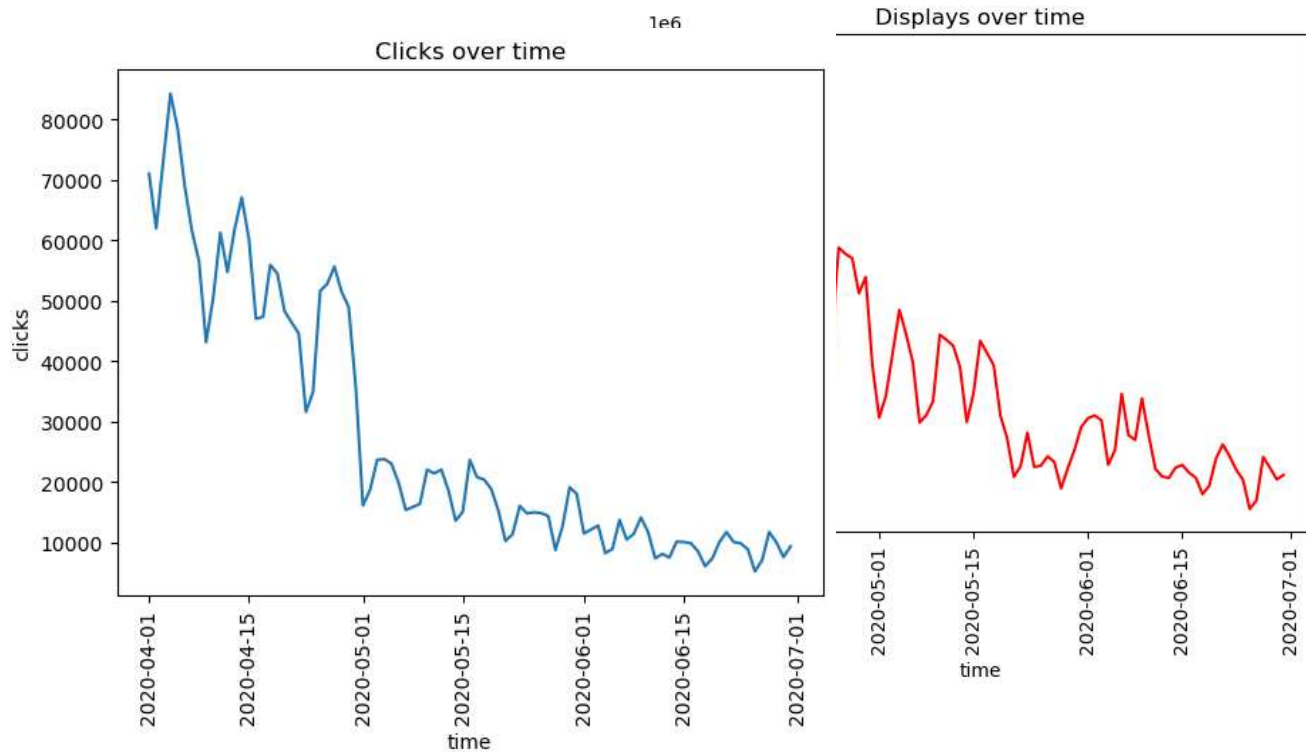
```
jkl       20853
```

```
abc         824
```

```
Name: post_click_conversions, dtype: int64
```

```
GHI has highest post click conversion rates
```

Can we identify any seasonal patterns or fluctuations in displays and clicks throughout the campaign period?



Clearly both the displays and clicks decrease with the time period. However, there are regular peaks in the graphs indicating week day type specific increase in the click and display

In general, the correlation of the clicks and displays are high. With higher clicks is higher displays



Is there a correlation between user engagement levels and the revenue generated?

Correlation between engagement and revenue: 0.175

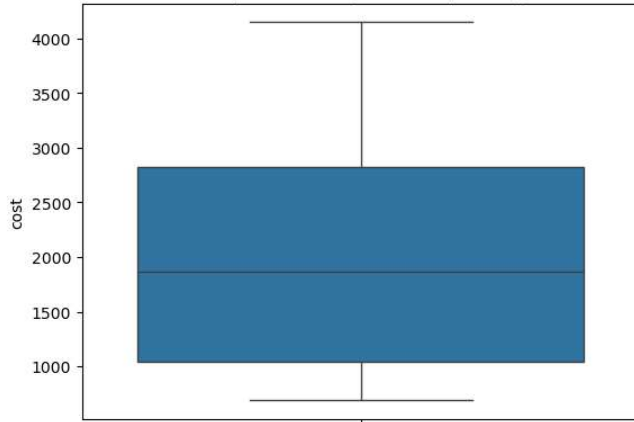
Since engagement is a string, the given mapping was used:

```
map({'Low': 1, 'Medium': 2, 'High': 3})
```

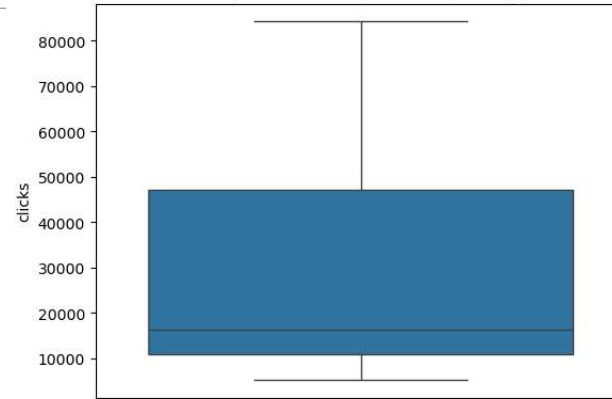
However it was found that the revenue generated is not very correlated with the engagement levels. That means higher engagement doesn't guarantee higher correlation

Are there any outliers in terms of cost, clicks, or revenue that warrant further investigation?

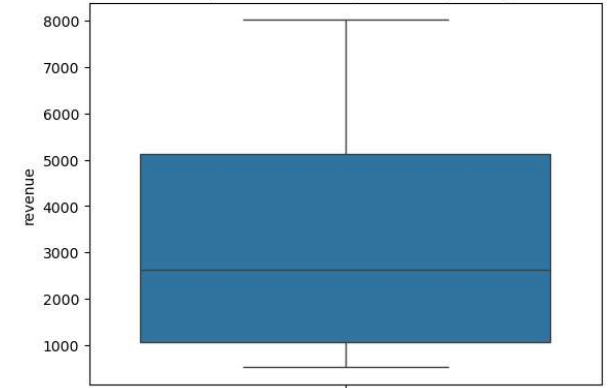
Box plot for cost(total sum per day)



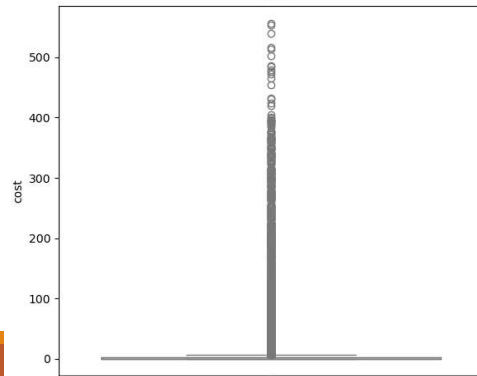
Box plot for clicks(total sum per day)



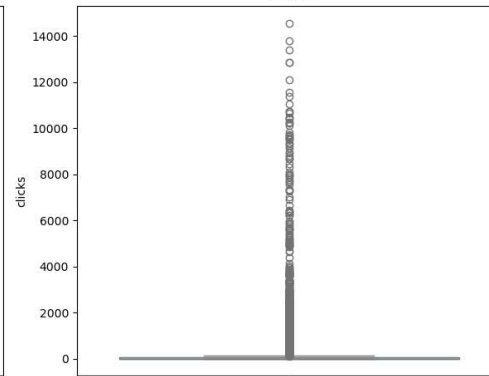
Box plot for revenue (total sum per day)



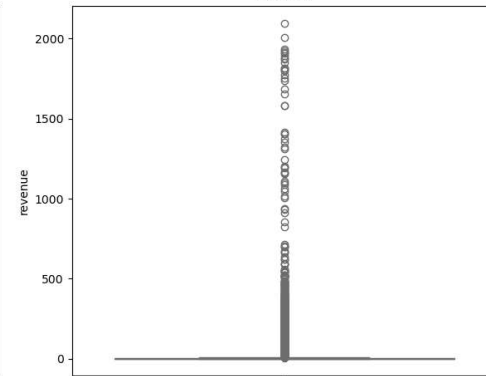
Cost



Clicks



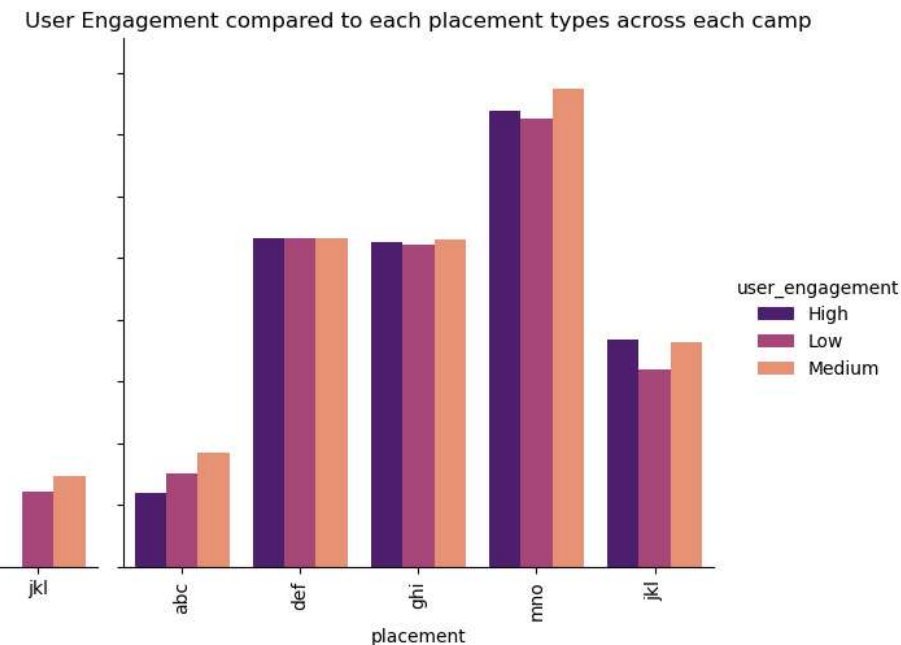
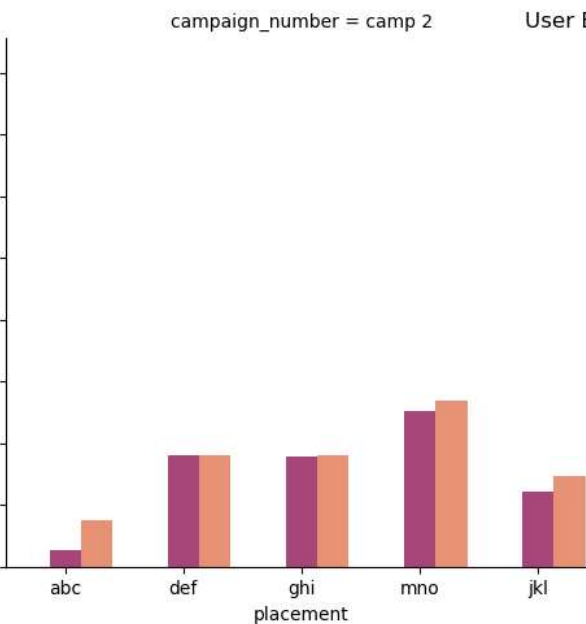
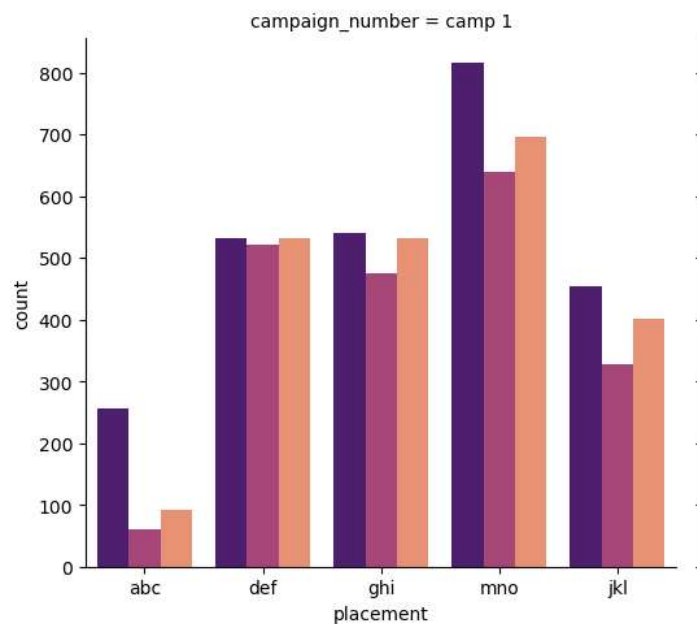
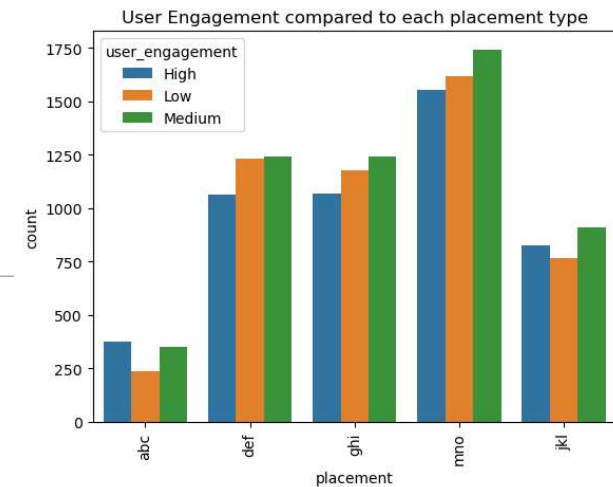
Revenue



The first three shows the box plot when the data set was date and the total sum of each parameter each day. We see that there are no outliers present in the data and it is fairly regular.

The bottom set of boxplots refer to the value of parameters taken entry by entry. In this case there are entries with 0 values and very high values. They show LARGE number of outliers but that doesn't have any meaning since at the end of the day total parameter per day counts.

How does the effectiveness of campaigns vary based on placement type?



Clearly Placement MNO has the most user engagement both in number and in type. The high and medium engagement is High in it

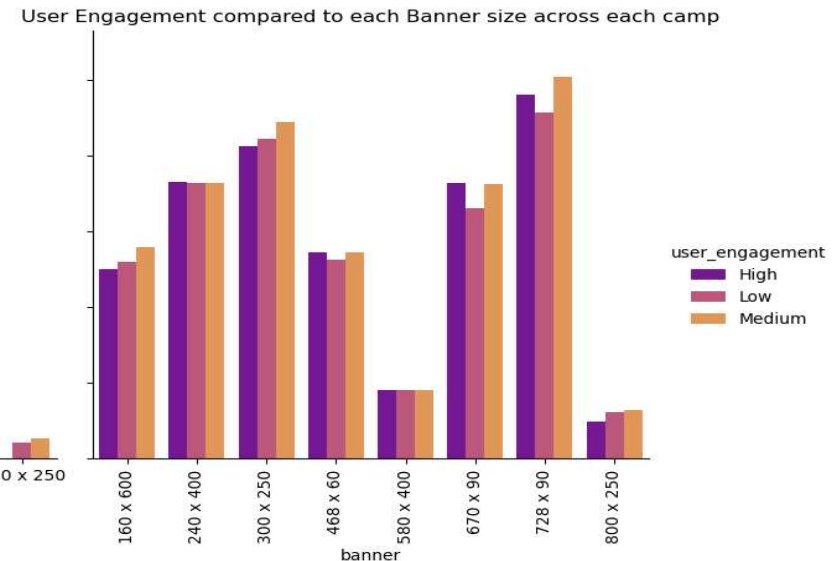
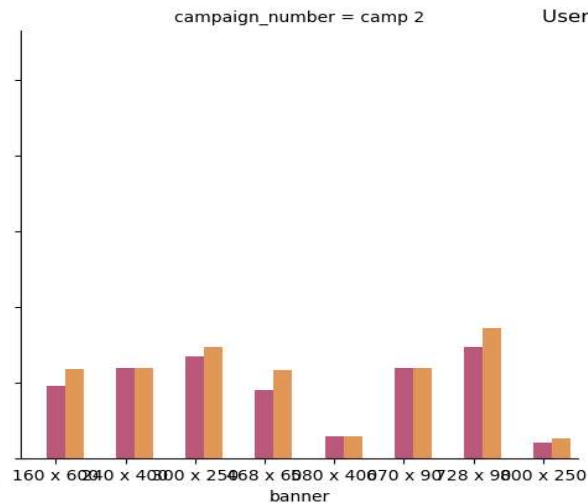
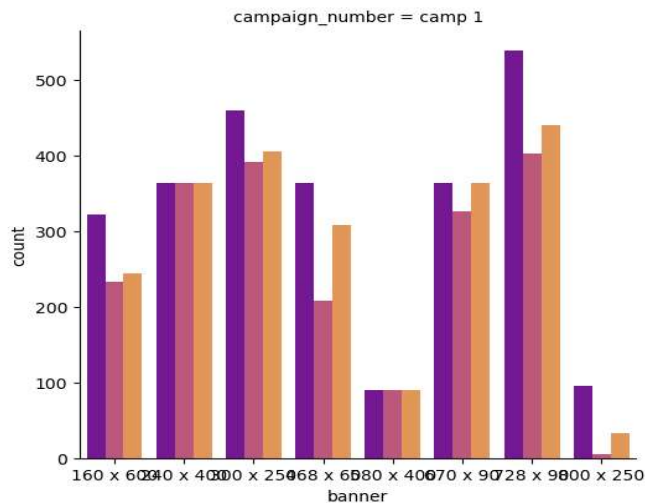
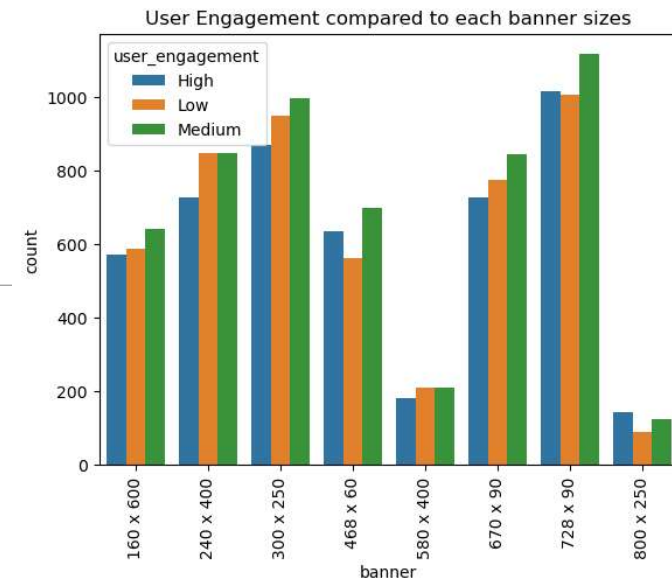
Comparing campaign number wise, in camp 2 there is NO HIGH engagement by any placement.

In camp 1 and 3, MNO has the highest engagement, in all three types.

DEF AND GHI performs very similarly to each other in all camps. Both are performing in a similar fashion

ABC is underperforming in number. HOWEVER, in camp A, ABC has HIGH engagement compared to the number of low and medium engagement. Relatively the engagement is better in Camp 1

How does the effectiveness of campaigns vary based on Banner sizes?



Camp 2 underperforms in this as well

Banner 728-90 high engagement compared to other banners.

160x600,240x400,300x250,670x90 etc also performs quite good

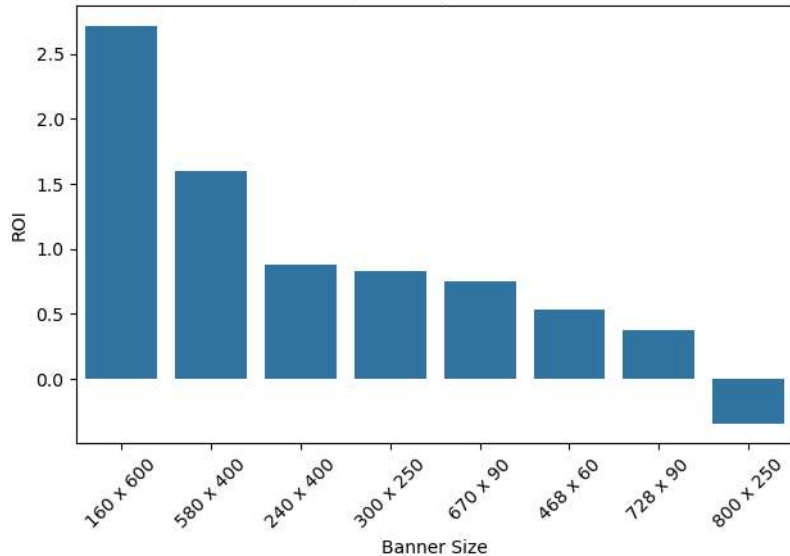
BANNER 800X250 and 580X400 performs very poorly

The overall engagement of any banner is comparatively poorer in camp 2

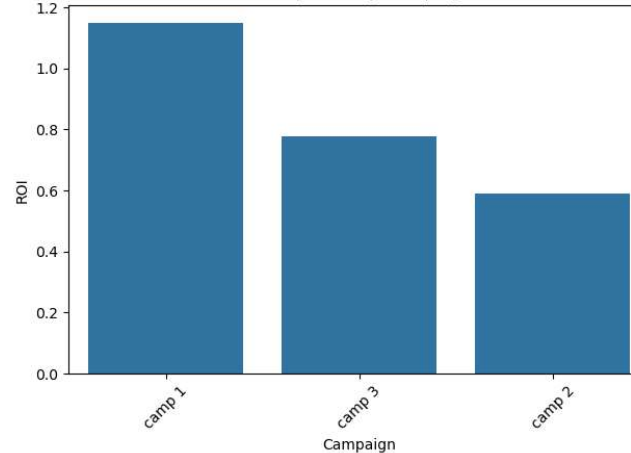
A solid orange horizontal bar spanning the width of the slide, located at the bottom.

Are there any specific campaigns or banner sizes that consistently outperform others in terms of ROI?

Average ROI by Banner Size



Average ROI by Campaign



```
Average ROI by Campaign:  
campaign_number  
camp 1      1.148795  
camp 3      0.776147  
camp 2      0.591237  
Name: ROI, dtype: float64
```

```
Average ROI by Banner Size:  
banner  
160 x 600    2.714339  
580 x 400    1.594362  
240 x 400    0.879907  
300 x 250    0.829696  
670 x 90     0.749633  
468 x 60     0.529702  
728 x 90     0.378855  
800 x 250    -0.339412  
Name: ROI, dtype: float64
```

$$\text{ROI} = (\text{REVENUE} - \text{COST}) / \text{COST}$$

In case of banners, we see a stark contrast. 160x600 banner gives huge ROI though the number, engagement etc are less compared to 728x90. 580x400 is also fairly good

However, though 728x90 has high user engagement, its ROI is less, i.e Revenue generated is not that good when cost to produce is considered.

800X250 has negative ROI indicating that cost to produce is higher than Revenue, i.e it is lossy.

In case of camps, clearly camp 1 gives the highest ROI followed by camp 3. Camp 2 is not that behind either

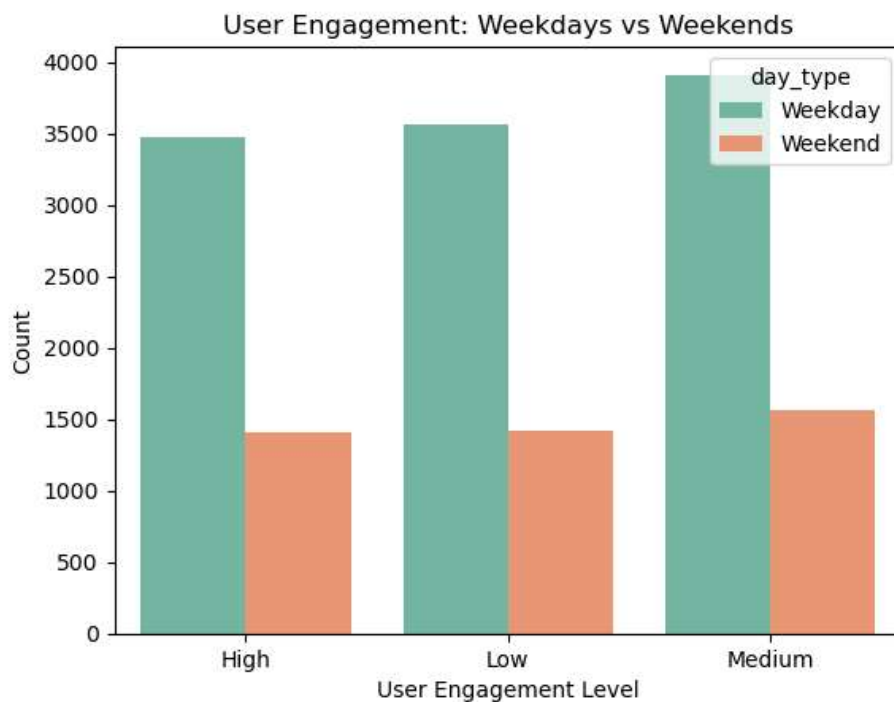


What is the distribution of post-click conversions across different placement types?

High in GHI and MNO and very low comparatively in ABC

```
Post click conversions per placment  
placement  
ghi      337063  
mno      263172  
def       29856  
jkl      20853  
abc         824
```

Are there any noticeable differences in user engagement levels between weekdays and weekends?



day_type	user_engagement	
Weekday	Medium	3911
	Low	3569
	High	3473
Weekend	Medium	1567
	Low	1419
	High	1405

Clearly the engagement is higher in weekdays. Medium engagement is more in both weekdays and weekends.

Weekdays are more engaging.

How does the cost per click (CPC) vary across different campaigns and banner sizes?

CPC IS Cost/Click

Clearly CPC is higher in camp 1 and low in camp 2

Among banners, 728X90 costs more per click while 800x250 has very less. However, its number of click itself is very low

```
campaign_number
camp 1      533.885775
camp 3      286.775424
camp 2       38.026495
Name: CPC, dtype: float64
banner
728 x 90      205.715482
240 x 400     177.676788
300 x 250     154.731958
670 x 90      124.988747
160 x 600     105.505623
468 x 60       52.855492
580 x 400      37.098954
800 x 250       0.114650
Name: CPC, dtype: float64
```

How does the Revenue per click (RPC) vary across different campaigns and banner sizes? (ANALYSED FOR FUN)

Even then the revenue of Camp 1 per click is higher.

728X90 has higher RPC too. So even though it costs more per click, it also revenues more per click!

```
campaign_number
camp 1      685.210624
camp 3      241.299361
camp 2       50.379077
Name: RPC, dtype: float64
banner
728 x 90      221.773457
240 x 400      217.363621
300 x 250      196.759169
670 x 90       133.987434
160 x 600      121.991336
580 x 400       55.614485
468 x 60       28.591260
800 x 250       0.808300
Name: RPC, dtype: float64
```


Are there any campaigns or placements that are particularly cost-effective in terms of generating post-click conversions?

Effectiveness is post-click-conversions per cost

Camp 1 is very cost effective since it generates very high post click conversions per cost.

MNO is very effective among placements though GHI is not very far behind. ABC performs rather less.

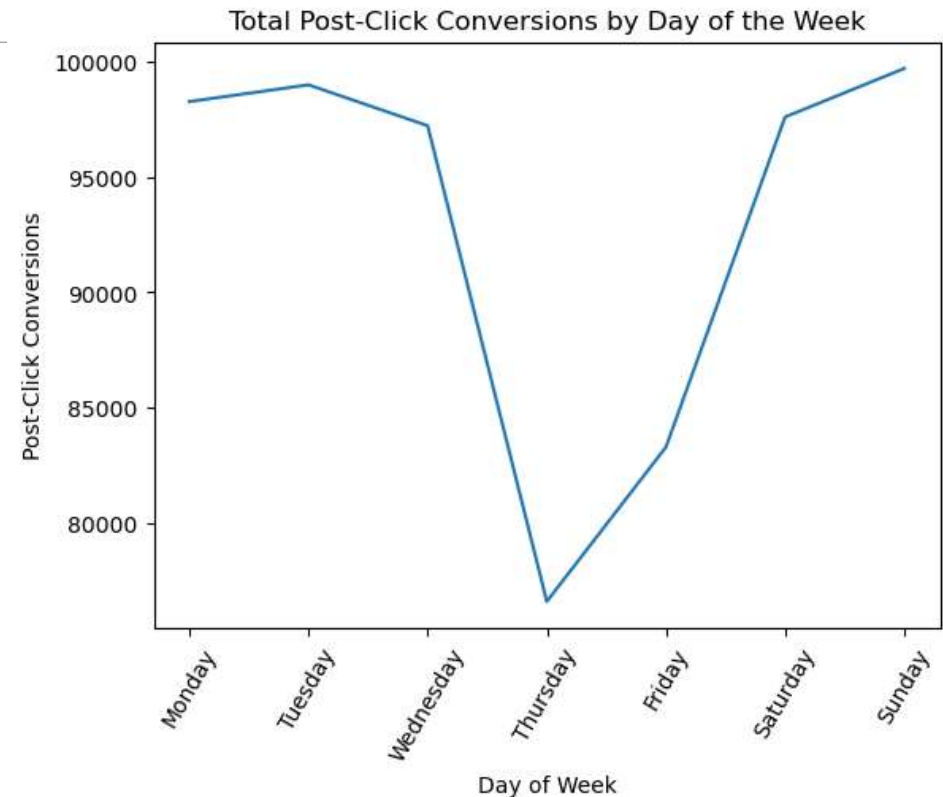
```
campaign_number
camp 1      19279.703458
camp 3       3263.433787
camp 2       1158.843227
Name: effectiveness, dtype: float64
placement
mno         6323.456913
ghi         5974.385208
def         5694.746762
jkl         4620.253366
abc         1089.138223
Name: effectiveness, dtype: float64
```

Can we identify any trends or patterns in post-click conversion rates based on the day of the week?

There is consistent and almost constant post click conversions from Saturday till Wednesday

HOWEVER, it decreases sharply on Thursday and slowly increases on Friday to reach the almost constant value on Saturday. There is a sharp dip only on Thursday. Sunday performs slightly better

day_name	
Sunday	99711
Tuesday	99001
Monday	98274
Saturday	97606
Wednesday	97227
Friday	83293
Thursday	76594

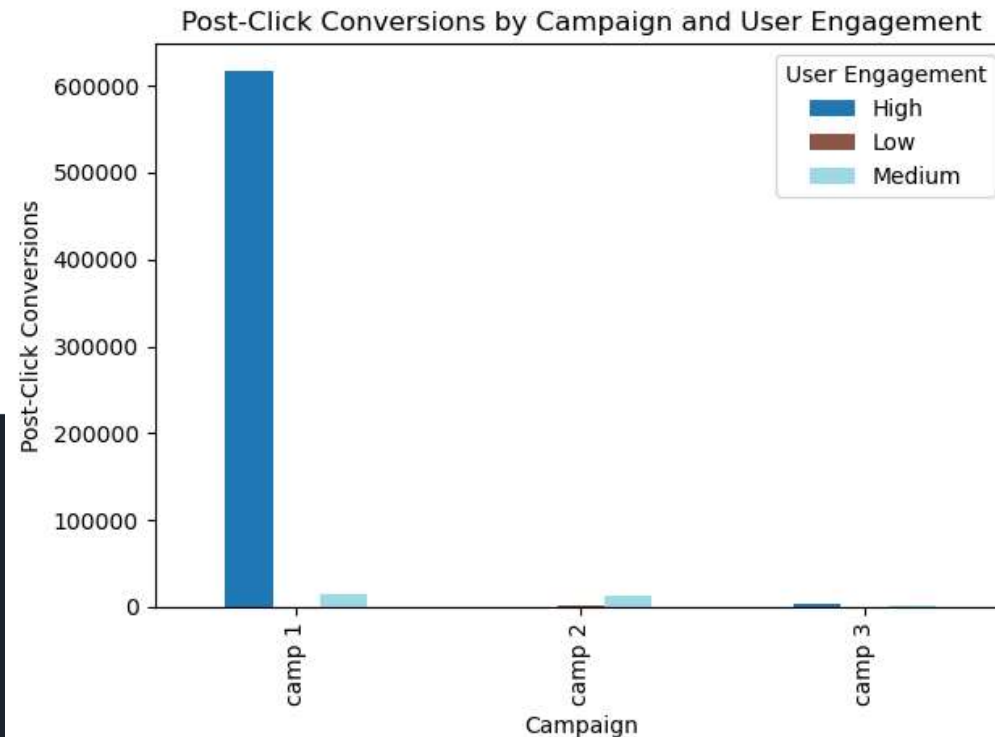


How does the effectiveness of campaigns vary throughout different user engagement types in terms of post-click conversions?

Camp 1 clearly is very effective in post click conversion number. It has a HIGH post conversions with HIGH user engagement.

Camp 2 and 3 is not even comparable to the User engagement in terms of post click conversion rates.

user_engagement	High	Low	Medium
campaign_number			
camp 1	617179	595	15249
camp 2	0	844	12923
camp 3	3681	150	1085



SUMMARY

- In camps, CAMP 1 performs the best
 - In banners 728x90 is the most preferable banner size
 - The data starts to decrease over time
 - Thursday has low post-click conversions
 - There is high correlation between many factors as shown in the correlation heat plot
- 

THANK YOU

